Resume Builder

Project 23 | Prompt Engineering

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1. Project overview

Goal: Use well-written prompts to make an AI chatbot build a complete, professional resume, plus a cover letter, for a fresher.

Sample candidate: Ananya Rao is a made-up fresher with a B.Tech in Computer Science who wants a Data Analyst job. All details in this document are samples for practice.

What this document contains: an information sheet, a reusable master prompt, six prompts with sample outputs (summary, project bullets, skills, full resume, tailoring to a job, cover letter), a before-and-after example, and tips.

Why prompts matter: An AI can only write about the candidate if the prompt gives it real facts. A prompt that includes the facts, the target job and the format gives a resume that is ready to use. A vague prompt gives a generic one.

2. The candidate information sheet

Before writing any prompt, collect the facts in one place. The AI will only use what is written here, so it cannot invent details.

Item	Details (sample)
Name	Ananya Rao
Target job	Data Analyst (fresher)
Education	B.Tech Computer Science, Sample Institute of Technology, 2022 to 2026, CGPA 8.2 out of 10
Skills	Python, Pandas, SQL, Power BI, Excel, Scikit-learn, Matplotlib
Project 1	Restaurant Sales Analysis: analysed 600 orders with Pandas, found top items, best months and payment trends, and built a dashboard with 4 charts
Project 2	Student Placement Prediction: trained Logistic Regression, Decision Tree and Random Forest models on 500 student records and compared their accuracy
Project 3	Blinkit Sales Dashboard: built a Power BI dashboard with cards, charts and a slicer to track sales and orders
Certification	Data Science with Python, Sample Training Institute
Soft skills	Teamwork, communication, problem solving

Item	Details (sample)
Contact	ananya.rao@example.com, +91 90000 00000

3. The master prompt for resumes

Copy this template, fill in the brackets, and paste it into any AI chatbot.

```
You are an expert resume writer who helps freshers get interview calls.
Task: Write [WHAT TO CREATE] for [TARGET JOB].
Context: Use only the facts below and do not invent anything.
[PASTE THE INFORMATION SHEET HERE]
Format: [LENGTH, STRUCTURE, NUMBER OF BULLETS].
Tone: [TONE WORDS].
Do not: [THINGS TO AVOID].
```

The most important line is **Use only the facts below and do not invent anything.** It stops the AI from adding fake jobs, fake numbers or skills the candidate does not have.

4. Six prompts with sample outputs

Prompt 1: Professional summary

Prompt:

```
You are an expert resume writer who helps freshers get interview calls. Write a
professional summary for a Data Analyst fresher. Context: Ananya Rao has a B.Tech in
Computer Science (CGPA 8.2) and skills in Python, Pandas, SQL, Power BI and Excel. She
built a sales analysis project, a placement prediction model and a Power BI dashboard.
Use only these facts. Format: 3 sentences, maximum 50 words, no first-person words like I
or my. Tone: confident and professional. Do not use the words passionate or hardworking.
```

Sample output:

> Computer Science graduate with hands-on experience in Python, SQL and Power BI. Built sales analysis, machine learning and dashboard projects that turn raw data into clear insights. Seeking a Data Analyst role to support data-driven decisions.

Why it works: The word limit keeps it short, the no-first-person rule matches resume style, and the Do not line removes overused words.

Prompt 2: Project bullet points

Prompt:

```
You are an expert resume writer. Write resume bullet points for the project below.
Context: Restaurant Sales Analysis. Analysed 600 orders with Pandas, found top items,
best months and payment trends, and built a dashboard with 4 charts. Use only these
facts. Format: exactly 3 bullet points, each starting with a strong action verb, each
under 20 words. Tone: clear and specific. Do not add numbers or tools that are not in the
facts.
```

Sample output:

- Analysed 600 restaurant orders using Python and Pandas to find top-selling items and best-performing months.
- Identified payment method trends to understand how customers prefer to pay.
- Built a 4-chart dashboard that presents sales insights in one clear view.

Why it works: Starting each bullet with an action verb makes the work sound active, and the facts-only rule keeps every claim true.

Prompt 3: Skills section

Prompt:

```
You are an expert resume writer. Organise the skills below into a resume skills section
for a Data Analyst fresher. Context: Python, Pandas, SQL, Power BI, Excel, Scikit-learn,
Matplotlib, teamwork, communication, problem solving. Format: three labelled lines named
Programming and Data, Tools and Visualisation, and Soft Skills. Keep every skill in only
one line. Do not add any skill that is not in the list.
```

Sample output:

- **Programming and Data:** Python, Pandas, SQL, Scikit-learn
- **Tools and Visualisation:** Power BI, Excel, Matplotlib
- **Soft Skills:** Teamwork, communication, problem solving

Why it works: Grouping skills makes them easy for a recruiter to scan in a few seconds.

Prompt 4: Full one-page resume

Prompt:

```
You are an expert resume writer who helps freshers get interview calls. Write a complete
one-page resume for a Data Analyst fresher. Context: use only the facts in the
information sheet below and do not invent anything. [Paste the full information sheet
from Section 2.] Format: plain text sections in this order: Header, Summary, Education,
Skills, Projects, Certification. Give each project 2 bullet points that start with an
action verb. Keep it ATS friendly with no tables or graphics. Tone: professional.
```

Sample output:

ANANYA RAO
ananya.rao@example.com | +91 90000 00000

SUMMARY
Computer Science graduate with hands-on experience in Python, SQL and Power BI. Built sales analysis, machine learning and dashboard projects that turn raw data into clear insights. Seeking a Data Analyst role.

EDUCATION
B.Tech, Computer Science, Sample Institute of Technology, 2022 to 2026, CGPA 8.2 / 10

SKILLS
Programming and Data: Python, Pandas, SQL, Scikit-learn
Tools and Visualisation: Power BI, Excel, Matplotlib
Soft Skills: Teamwork, communication, problem solving

PROJECTS

Restaurant Sales Analysis

- Analysed 600 orders using Pandas to find top items, best months and payment trends.
- Built a 4-chart dashboard to present the insights clearly.

Student Placement Prediction

- Trained Logistic Regression, Decision Tree and Random Forest models on 500 student records.
- Compared model accuracy to select the best-performing model.

Blinkit Sales Dashboard

- Built a Power BI dashboard with cards, charts and a slicer to track sales and orders.
- Used measures to calculate total sales, total orders and average order value.

CERTIFICATION
Data Science with Python, Sample Training Institute

Why it works: The fixed section order and the ATS-friendly rule give a clean resume that applicant tracking software can read.

Prompt 5: Tailor the resume to a job description

Prompt:

```
You are an expert resume coach. Compare the resume summary and skills below with the job
description and help me tailor them. Job description: Data Analyst Intern. Requirements:
SQL, Excel, Power BI dashboards, data cleaning, and the ability to explain insights to
the team. Resume summary: Computer Science graduate with hands-on experience in Python,
SQL and Power BI. Skills: Python, Pandas, SQL, Power BI, Excel, Scikit-learn, Matplotlib.
Format: first list the keywords from the job description that the resume already has,
then the keywords that are missing, then rewrite the summary in 40 words or less using
the matching keywords. Do not claim any skill that is not in my skills list.
```

Sample output:

> **Keywords already in the resume:** SQL, Excel, Power BI, dashboards.
>
> **Keywords missing:** data cleaning, explaining insights to a team.
>
> **Note:** Add data cleaning only if it is true. The Restaurant Sales Analysis project included checking and cleaning data, so it can be added to that project.
>
> **Tailored summary:** Computer Science graduate skilled in SQL, Excel and Power BI dashboards. Experienced in cleaning and analysing data with Python and Pandas, and presenting insights clearly. Seeking a Data Analyst Intern role.

Why it works: Matching the words in the job description helps the resume pass automatic screening, and the last rule keeps it honest.

Prompt 6: Cover letter

Prompt:

```
You are an expert career coach. Write a short cover letter for Ananya Rao applying for a
Data Analyst Intern job at a company called BrightPath Analytics. Context: she has a
B.Tech in Computer Science, skills in SQL, Excel, Power BI and Python, and projects in
sales analysis and a Power BI dashboard. Use only these facts. Format: 120 to 150 words,
3 short paragraphs (introduction, strengths, closing with a request for an interview).
Tone: polite, confident and enthusiastic. Do not use made-up achievements.
```

Sample output:

> Dear Hiring Manager,
>
> I am writing to apply for the Data Analyst Intern position at BrightPath Analytics. I am a Computer Science graduate with a strong interest in turning data into useful insights.
>
> During my studies I built a restaurant sales analysis using Python and Pandas, and a Power BI dashboard that tracks sales and orders. These projects taught me to clean data, find patterns and present results clearly. I am also comfortable with SQL and Excel.
>
> I would be glad to bring these skills to your team and keep learning from your analysts. Thank you for your time. I would welcome the chance to discuss my application in an interview.
>
> Sincerely, Ananya Rao

Why it works: The three-paragraph format and the facts-only rule give a short, honest letter that is easy to personalise.

5. Improving a weak prompt (before and after)

Weak prompt:

```
Make a resume for me.
```

Typical result: A generic template with made-up skills and jobs, such as five years of experience at a company the candidate never worked for. This is dangerous, because a resume must be true.

Improved prompt: Prompt 4 above.

Result: A one-page resume that uses only the real facts, follows a clear order and is ready for applicant tracking software.

What was added	Why it helps
Role	Gives the AI the voice of an expert resume writer
Target job	Chooses the right keywords and skills
Facts only	Stops the AI from inventing experience
Fixed section order	Gives a clean, standard layout
Bullets with action verbs	Makes the achievements sound active
ATS friendly	Avoids tables and graphics that software cannot read

6. Tips for better resume prompts

- **Collect your facts first.** Fill in the information sheet before writing any prompt.
- **Always say do not invent anything.** A resume must be true, and you must be able to explain every line in an interview.
- **Name the target job.** A resume for a Data Analyst is different from one for a Web Developer.
- **Use action verbs.** Built, analysed, designed and compared sound stronger than worked on.
- **Keep it to one page** for a fresher, with clear headings.
- **Tailor it for each job.** Use Prompt 5 to match the keywords in each job description.
- **Check before sending.** Read every line, check spelling, and remove anything you cannot explain.

7. Conclusion

This project shows that an AI can build a strong resume, but only when the prompt supplies real facts, a target job and a clear format. With one information sheet and one master prompt, the same candidate can get a summary, project bullets, a skills section, a full resume, a job-tailored version and a cover letter in a few minutes.